

3.Rectifiers (without and with c-filter)

A. HALF -WAVE RECTIFIER WITH AND WITHOUT FILTER

AIM: To examine the input and output waveforms of half wave Rectifier and also Calculate its load regulation and ripple factor.

1. With Filter
2. Without Filter

APPARATUS:

Digital Multimeter	- 1No.
Transformer (6V-0-6V)	- 1No.
Diode, 1N4007	- 1No.
Capacitor 100 μ f/470 μ f	- 1No.
Decade Resistance Box	- 1No.
Breadboard	
CRO and CRO probes	
Connecting wires	

THEORY:

In Half Wave Rectification, When AC supply is applied at the input, only Positive Half Cycle appears across the load whereas, the negative Half Cycle is suppressed. How this can be explained as follows:

During positive half-cycle of the input voltage, the diode D1 is in forward bias and conducts through the load resistor R_L . Hence the current produces an output voltage